

Opuntia ficus-indica (Prickly Pear Cactus) Fiber: Efficacy and Safety in Weight Management (Literature Review)

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ABSTRACT

Background:

Obesity is a global health epidemic associated with increased risks of type 2 diabetes, cardiovascular disease, and other comorbidities. Reducing dietary fat absorption is one strategy for weight management. Opuntia ficus-indica (OFI, prickly pear cactus) is rich in soluble and insoluble fibers that may bind fat and reduce its absorption, thereby assisting weight control. This review summarizes human clinical evidence on the efficacy and safety of OFI-derived fiber in weight management.

Methods:

Peer-reviewed clinical studies investigating OFI cactus fiber (often formulated as the proprietary product Litramine™ or NeOpuntia®) for weight loss or fat excretion were reviewed. Key outcomes included body weight change, body composition, fecal fat excretion, metabolic parameters, and adverse events. Supporting mechanistic and preclinical data are also discussed.

Results:

Multiple randomized controlled trials indicate that cactus fiber supplementation can promote modest but significant weight loss in overweight individuals when used alongside diet. In a 12-week trial, subjects receiving 3 g/day OFI fiber lost significantly more weight than those on placebo (approximately 3.8 kg vs 1.4 kg, $p < 0.001$) with no serious side effects (Grube *et al.*, 2013). Another placebo-controlled study showed that cactus fiber increased fecal fat excretion by over 3-fold compared to placebo (15.8% vs 4.6% of dietary fat, $p < 0.001$), supporting its mechanism of action (Uebelhack *et al.*, 2014). A 24-week follow-up study suggested that continued use of OFI fiber helped maintain prior weight loss, with the active group sustaining weight loss while the placebo group tended to regain weight (Chong *et al.*, 2013). OFI fiber was well tolerated across studies, with gastrointestinal symptoms mild and transient.

Discussion:

Cactus fiber from *Opuntia ficus-indica* appears to be an effective and safe adjunct for weight management. Its soluble fiber component binds dietary fat in the gut, reducing fat absorption and calorie intake. Clinical trials demonstrate meaningful improvements in weight loss and metabolic outcomes with OFI fiber, without the gastrointestinal side effects or safety concerns associated with pharmaceutical fat-blockers like orlistat. However, variability in individual responses and product formulations exists. More research will shed more light on long-term efficacy, optimal dosing, and effects on obesity-related comorbidities. Current evidence supports OFI cactus fiber as a beneficial tool for weight control as part of a balanced diet and lifestyle.

Subjects: Dietary fats, Nutrition, Weight Management, **Keywords:** Obesity, Intervention

INTRODUCTION

Overweight and obesity have reached pandemic proportions worldwide. As of 2013, more than 1.4 billion adults were overweight, and over 500 million were obese, according to the World Health Organization (World Health Organization, 2013). This excess adiposity is associated with numerous chronic health issues, including type 2 diabetes, dyslipidemia, hypertension, cardiovascular disease, and certain cancers (Perez, 2013; Agborsangaya *et al.*, 2013). Obesity also increases the odds of multimorbidity (the coexistence of multiple chronic conditions) and reduces quality of life (Agborsangaya *et al.*, 2013; Cawley & Meyerhoefer, 2012). The economic and healthcare burdens of obesity are substantial (Cawley & Meyerhoefer, 2012).

High dietary fat intake is believed to be a major contributor to obesity. Fat is energy-dense, and studies suggest that calories derived from fat promote weight gain more than the same caloric amount from carbohydrates or protein (Rolls, 2000). In populations, increased fat consumption correlates with higher obesity prevalence (Elmadfa *et al.*, 2009). Therefore, reducing fat intake or blocking fat absorption are logical strategies for weight management. One pharmacological approach, the pancreatic lipase inhibitor orlistat, prevents about 25–30% of dietary fat from being absorbed. Orlistat is effective for weight loss, but it commonly causes unpleasant gastrointestinal side effects such as oily stool, fecal urgency, and flatulence (Rucker *et al.*, 2007). More serious concerns have also been raised, including rare instances of liver injury (Douglas *et al.*, 2013).

As an alternative to drugs, natural fibers have attracted attention for weight control. Dietary fiber can increase satiety, reduce overall energy absorption, and promote fat excretion (Baer *et al.*, 1997; Tucker & Thomas, 2009). Prospective studies show that higher fiber intake is associated with less weight gain over time (Tucker & Thomas, 2009). However, different fibers vary in their fat-binding capacities and weight management

effects (Kristensen *et al.*, 2012). Among these, fiber from the prickly pear cactus (*Opuntia ficus-indica*, OFI) has shown promise as a weight-loss supplement.

In this literature review, we examine the evidence for OFI-derived fiber in aiding weight management. We focus on human clinical trials evaluating an OFI fiber complex (commonly marketed as Litramine™ IQP-G-002AS or NeOpuntia®) for weight loss, fat excretion, and weight maintenance. We also discuss the mechanism by which cactus fiber interacts with dietary fat, and we summarize safety findings. The goal is to provide a comprehensive overview of how *Opuntia ficus-indica* fiber can contribute to obesity management in a clinical context.

Background

Opuntia ficus-indica (L.) Mill., commonly known as prickly pear or nopal cactus, is a succulent plant native to Mexico that has been cultivated and naturalized in arid regions worldwide, including the Mediterranean and North Africa. The young cactus pads (cladodes) and fruits have a long history of use as food and traditional medicine. The cactus pads are especially rich in dietary fiber, comprising approximately 40–50% of their dry weight (Saenz, 1997). This fiber content includes both soluble fibers (such as mucilage, pectin, gums, and hemicellulose) and insoluble fibers (primarily cellulose and hemicellulose) (Hernandez-Urbiola *et al.*, 2011). The proportion of soluble to insoluble fiber in OFI can vary with the plant's age: as the cladodes mature, insoluble fiber content increases while soluble fiber decreases (Hernandez-Urbiola *et al.*, 2011; Hernandez-Urbiola, 2010). Other factors such as cactus variety, growing conditions, and processing methods can also influence the fiber composition and, consequently, the health effects of OFI products (Peña-Valdivia *et al.*, 2012).

OFI has been studied for various health benefits. Its fiber and bioactive compounds (e.g., antioxidants in the fruits) have been reported to improve blood glucose control and blood lipid profiles in some studies (Godard *et al.*, 2010; Frati-Munari *et al.*, 1989; Frati-Munari *et al.*, 1983). For example, supplementation with OFI extracts (such as OpunDia, derived from cactus pads) acutely lowered blood glucose in pre-diabetic individuals and appeared to be safe over the longer term (Godard *et al.*, 2010). Some early clinical investigations in Mexico suggested that consuming nopal cactus could reduce serum cholesterol and glucose, and even modestly decrease body weight (Frati-Munari *et al.*, 1983). However, findings across studies have been inconsistent. Other trials did not find significant metabolic improvements or weight changes with nopal, possibly due to differences in preparation, dosage, or study design (Frati-Munari *et al.*, 1992). Thus, while the potential health benefits of *Opuntia ficus-indica* have been recognized, consistent clinical evidence was lacking until more recent controlled trials.

A key functional property of prickly pear cactus is its ability to bind dietary fat in the gastrointestinal tract. The soluble fiber in OFI forms a gel-like matrix that can entrap lipids

and bile acids, thereby reducing fat digestion and absorption. In vitro tests have demonstrated that a specific powdered OFI fiber product (NeOpuntia®) binds a significant amount of fat under gastrointestinal conditions (Izani *et al.*, 2009). In a dynamic GI model, OFI fiber reduced the availability of fat for absorption, supporting the hypothesis that it can function as a fat blocker (Granato, 2005). A small pilot clinical study with healthy volunteers provided initial proof of concept: consuming 4.8 g of OFI fiber per day led to about 27% more fat being excreted in feces compared to placebo, indicating that the fiber was indeed preventing fat absorption (Bachmann, 2010). This fat-binding mechanism is analogous to – though generally milder than – that of orlistat, but coming from a natural fiber source which might avoid the intense side effects.

To optimize and standardize the fat-binding capacity of cactus fiber, a proprietary complex known as Litramine™ (IQP-G-002AS) was developed. Litramine is derived from *Opuntia ficus-indica* cladode fiber that is purified and blended with a soluble fiber from Acacia (gum Arabic), then granulated with a small amount of cyclodextrin to enhance its action. This formulation yields a consistent fat-binding performance in vitro (Grube *et al.*, 2013). Litramine has been incorporated into over-the-counter weight management products (for example, marketed in Europe under brand names like XLS-Medical® Fat Binder). The following sections review the clinical efficacy of this OFI fiber complex in controlled human studies and evaluate its safety profile.

Clinical Evidence

Randomized Controlled Trials of Cactus Fiber

Several randomized, placebo-controlled trials have investigated the effects of *Opuntia ficus-indica* fiber on weight loss and related outcomes in overweight adults. Table 1 provides an overview of the key clinical studies and their findings. The trials vary in duration from 45 days to 24 weeks and involve individuals who are overweight or mildly to moderately obese (generally with BMI between ~25 and 35 kg/m²).

One of the foundational studies was a 3-month double-blind RCT conducted in Germany to test the efficacy of Litramine (cactus fiber complex) in promoting weight loss (Grube *et al.*, 2013). In this trial (n=125), participants were given either Litramine (3 g daily, in divided doses of two tablets after each main meal) or a matching placebo, alongside a calorie-reduced diet. After 12 weeks, the Litramine group had a significantly greater reduction in body weight compared to the placebo group. On average, subjects taking the cactus fiber lost about 3.8 kg, whereas those on placebo lost about 1.4 kg ($p<0.001$) (Grube *et al.*, 2013). In other words, the fiber-supplemented group achieved roughly three times more weight loss than diet alone. Additionally, a higher proportion of the fiber group met clinically meaningful weight loss thresholds (such as $\geq 5\%$ of initial body weight). Importantly, no serious adverse events were reported, and tolerability was good, with gastrointestinal complaints similar to placebo in frequency and mild in nature (Grube *et al.*, 2013).

Another pivotal study focused on the fat-binding action of the cactus fiber. Uebelhack *et al.* (2014) conducted a crossover trial in healthy volunteers to directly measure fecal fat excretion – an indicator of fat malabsorption – during fiber supplementation. Twenty subjects followed a controlled diet and were randomized to receive either cactus fiber tablets (Litramine, 2 tablets three times daily, providing a total of 3 g fiber per day) or placebo for a brief treatment period, then crossed over to the other arm after a washout. Even over this short period (45 days with each condition, including washout), the effects were dramatic: the proportion of dietary fat excreted in feces was $15.79\% \pm 5.8\%$ with cactus fiber vs $4.56\% \pm 3.1\%$ with placebo – a statistically significant difference ($p < 0.001$) (Uebelhack *et al.*, 2014). In essence, the cactus fiber caused subjects to lose about one-sixth of their dietary fat intake in stools, compared to about one-twentieth on placebo. Despite this malabsorptive effect, the fiber was well tolerated; participants did not report adverse gastrointestinal symptoms beyond what placebo caused (Uebelhack *et al.*, 2014). This trial elegantly demonstrated the mechanism by which OFI fiber can aid in weight management, by reducing effective calorie absorption. While weight change was not a primary endpoint in this short study, the authors noted that chronic use of such a fiber could translate into weight loss, supporting the findings of longer trials like Grube *et al.* (2013).

After achieving weight loss, preventing weight regain is a major challenge. A follow-up study examined Litramine's role in weight maintenance (Chong *et al.*, 2013). In that 24-week investigation, subjects who had recently lost weight were either continued on Litramine or given placebo to observe weight trajectory. Over the extended period, those who took the cactus fiber complex were better able to maintain their weight loss: the Litramine group sustained their reduced weight or continued to lose slightly, whereas the placebo group tended to regain weight that had been lost earlier. By the end of the trial, the difference in weight maintenance between groups was significant, indicating that OFI fiber can help prevent the typical rebound weight gain post-diet (Chong *et al.*, 2013). This finding is valuable, as weight maintenance is often overlooked in supplement research. It suggests that ongoing intake of cactus fiber could assist not only in losing weight but also in keeping it off. Additionally, this extended use confirmed the product's safety, with no indication of systemic side effects or nutrient deficiencies over 6 months.

Other supporting clinical evidence comes from complementary studies. For example, Godard *et al.* (2010) showed that OFI pad extract (OpunDia) improved postprandial blood glucose without adverse effects, which may be relevant for overweight individuals with impaired glucose tolerance, although that study did not primarily measure weight change. Historical studies from the 1980s and 1990s in Mexico observed some weight reduction and metabolic improvements with cactus consumption in diabetic patients (Fрати-Munari *et al.*, 1983; Frати-Munari *et al.*, 1992). However, these older studies had mixed outcomes and methodological limitations. The modern trials with standardized OFI fiber (as discussed above) provide more robust and consistent evidence for its efficacy in weight management.

Table 1. Summary of Human Clinical Studies on Opuntia ficus-indica (Cactus) Fiber for Weight Management

Reference (Author, Year)	Study Design/Duration	Subjects (Population)	Purpose/ Endpoints	Preparation/ Dose (OFI Fiber)	Key Results
Grube et al., 2013	Randomized, double-blind, placebo-controlled trial; 12 weeks	n=125 overweight/obese adults (BMI 25–35)	Weight loss efficacy	Litramine™ cactus fiber, 3 g/day (2 tablets after each meal)	↓ Body weight vs placebo (–3.8 kg vs –1.4 kg, $p < 0.001$); greater % of subjects lost ≥5% weight; no serious adverse events.
Uebelhack et al., 2014	Randomized, double-blind, placebo-controlled crossover trial; ~45 days per arm	n=20 healthy adults	Fat binding (fecal fat excretion)	Litramine™ cactus fiber, 3 g/day (2 tablets thrice daily with meals)	↑ Fecal fat excretion vs placebo (15.8% vs 4.6% of dietary fat, $p < 0.001$); no adverse gastrointestinal effects reported.
Chong et al., 2013	Randomized, placebo-controlled trial; 24 weeks (12 weeks double-blind + 12 weeks open extension)	n=60 (approx.) overweight adults (after prior weight loss)	Weight loss maintenance	Litramine™ cactus fiber, 3 g/day (tablets)	Active group maintained weight loss (continued slight loss of ~0.6 kg) over 24 weeks, whereas placebo group regained weight (+~2.5 kg); significant between-group difference in weight maintenance. Fiber was well tolerated.
Reference (Author, date)	Study Design/Duration	Subjects	Purpose	Preparation/ Dose	Main Results
Thom, 2000	RPCT, 12 weeks	n = 40 (BMI 28–39)	weight loss	400 mg Phaseolamin® 3× after meals, total 1200 mg/day (other ingredients: inulin & Garcinia extract)	↓ body weight, BMI & % body fat in active group ($p < 0.05$), no effect in placebo group; no between-group analysis
Erner, 2003	RPCT, 12 weeks, then open 12 weeks	n = 54 (BMI 24–36)	weight loss	Thera-Slim: 1000 mg Phase 2 before 2 meals, total 2000 mg/day	trend toward ↓ body weight, 3× decrease in triglycerides; no between-group analysis
Rothacker, 2003	RPCT, 12 weeks	n = 88 (BMI 24–32)	weight loss	StarchAway chews: 1000 mg Phase 2 before 3 meals, total 3000 mg/day	↓ body weight compared to placebo ($p < 0.05$)

Udani, 2004	RPCT, 8 weeks	n = 39 (BMI 30–43)	weight loss	Phase 2 1500 mg 2×, 3000 mg/day	↓ body weight compared to placebo (ns); ↓ triglycerides (ns)
Koike, 2005	Open, 8 weeks	n = 10 (BMI 23–30)	weight loss	3 capsules Phaseolamin 1600 diet 2× daily; 750 mg Phase 2 daily	↓ body weight ($p = 0.002$), calorie intake, BMI, triglycerides & HDL (all $p < 0.05$)
Osorio, 2005	Open, 30 days	n = 39 (overweight & obese)	weight loss	Precarb capsules: 1000 mg Phase 3 with meals, total 3000 mg/day	↓ body weight & ↓ waist-to- hip ratio over time (both $p < 0.001$)
Celleno, 2007	RPCT, 30 days	n = 60 (BMI ~26)	weight loss	Phase 2 + chromium; 445 mg extract daily	↓ body weight ($p < 0.001$), BMI, body fat (both $p < 0.01$)
Vinson, 2009	PCT, crossover, single dose	Part 1: n = 11; Part 2: n = 7	plasma glucose	Phase 2 mixed with margarine or gravy; 750 or 1500 mg	↓ AUC post-prandial blood glucose; higher dose ($p < 0.05$)
Udani, 2009	RPC, crossover, single dose	n = 13 (BMI 18–25)	plasma glucose	Phase 2 capsules or mixed with butter; 1500, 2000, 3000 mg	↓ AUC post-prandial blood glucose; 3000 mg with butter ($p < 0.05$)
Wu, 2010	RPCT, 60 days	n = 101 (BMI 25–40)	weight loss	Phase 2; 1000 mg 3× daily	↓ body weight, waist circumference (both $p < 0.01$)

Note: ↓ indicates a decrease; ↑ indicates an increase. All changes described are relative to baseline or placebo. No serious adverse events (AEs) were attributed to the cactus fiber in any study. Mild gastrointestinal symptoms (e.g., bloating, flatulence) were occasionally reported, but their incidence did not significantly differ from placebo.

Mechanism and Comparative Studies

The efficacy of OFI cactus fiber in these trials is attributable to its ability to bind fats in the gastrointestinal tract, thereby reducing calorie absorption. This mechanism has also been observed with other fiber supplements and fat-binding agents. For instance, α -Cyclodextrin, a soluble fiber derivative, can bind dietary fat in a roughly 1:1 ratio by weight and has shown some promise as a weight management aid (Jen et al., 2013). Chitosan, a fiber-like substance from crustacean shells, similarly increases fecal fat excretion (Gades & Stern, 2003). Compared to these, the cactus fiber complex Litramine is notable for

having demonstrated both substantial fat binding *and* clinically significant weight loss in controlled trials.

In general, higher fiber intake contributes to weight control not only by blocking some fat calories but also by promoting satiety and stabilizing blood sugar. Soluble fibers like pectin, β -glucan, and those in OFI delay gastric emptying and can attenuate postprandial glucose spikes (Bourdon *et al.*, 1999; Sanaka *et al.*, 2007). This may reduce appetite and subsequent calorie intake. The combination of these effects – reduced absorption of fats and improved appetite regulation – likely underlies the weight loss observed with cactus fiber supplementation.

Safety and Tolerability

All peer-reviewed studies to date indicate that *Opuntia ficus-indica* fiber is safe and well tolerated when taken at recommended dosages (often around 3 g per day). Unlike pharmaceutical fat blockers, cactus fiber does not typically cause severe gastrointestinal side effects such as oily leakage or incontinence. In the clinical trials reviewed, the incidence of GI symptoms (like bloating, gas, or mild constipation) was low and usually comparable between the fiber and placebo groups (Grube *et al.*, 2013; Uebelhack *et al.*, 2014). Any digestive discomfort tends to be transient, often resolving as the body adapts to increased fiber intake. This contrasts with orlistat, where up to 15–30% of users experience notable GI distress (Rucker *et al.*, 2007).

No adverse changes in fat-soluble vitamin levels or other laboratory safety parameters were reported with cactus fiber in these studies. Litramine's manufacturer recommends taking a multivitamin at a different time of day as a precaution (since theoretically any fat-binding agent could modestly reduce fat-soluble vitamin absorption), but clinical data have not shown a deficiency or malabsorption of vitamins with use. The product's safety was further supported by a toxicological assessment in animals and an expert panel review, which found no toxic effects even at very high doses and concluded the fiber complex is safe for human consumption up to 10 g per day (Granato, 2005; Bachmann, 2010).

Cactus pads have been eaten traditionally for centuries, and aside from their fiber, they contain minimal anti-nutritional factors. One point of consideration: raw prickly pear pads have some tannins and oxalates, but these are not present in significant amounts in the processed fiber supplements. Also, unlike some legumes, cactus does not contain lectins or compounds that could cause systemic toxicity; any minor components are removed or inactivated during the production of the supplement. Overall, the safety profile of OFI fiber is excellent, making it a favorable option for long-term use in weight management.

DISCUSSION

The evidence compiled in this review suggests that *Opuntia ficus-indica* fiber can serve as an effective adjunct in the management of overweight and obesity. When used alongside dietary changes, the cactus fiber produces statistically significant and clinically meaningful weight loss in overweight individuals, on the order of 1.5–2.5 kg more than placebo over a few months (Grube et al., 2013). This magnitude of effect is comparable to some other non-pharmacological interventions and better than what has been seen with certain over-the-counter weight-loss supplements.

A crucial advantage of the OFI fiber complex is its mechanistic basis: by binding fat in the gut, it directly reduces caloric absorption. The 2014 trial by Uebelhack et al. elegantly confirmed that the product increases fecal fat output (Uebelhack et al., 2014), validating the physiological mechanism. Unlike stimulants or appetite suppressants, cactus fiber works locally in the intestines and thus has minimal systemic effect. This local action likely contributes to its strong safety profile.

Another notable finding is the benefit of cactus fiber for weight maintenance. Many individuals can lose weight through intensive diets but struggle to maintain the loss. The fact that continued use of OFI fiber helped prevent weight regain over 6 months (Chong et al., 2013) is encouraging. It indicates the fiber can be a useful tool not just for initial weight loss but also for the longer haul of weight control. This may be due to the fiber continuing to eliminate some calories from the diet and possibly helping users stick to their diet by mitigating hunger (through prolonged gastric emptying and improved satiety).

Comparatively, the weight loss achieved with cactus fiber is modest (a few kilograms) and certainly not a standalone solution for obesity. However, in a field where many purported supplements have little to no evidence, cactus fiber stands out as one of the few with multiple RCTs backing its efficacy. It offers a compromise between doing nothing and taking medications: a natural, low-risk approach that can augment lifestyle efforts. For individuals who are overweight but not ready or eligible for medications, or for those aiming to avoid pharmaceuticals, OFI fiber supplements provide a viable option.

Of course, there are limitations to consider. Most trials were relatively short (3 months is the longest double-blind phase). Long-term effects beyond six months remain to be fully studied. It is unclear if the body might adapt over time, potentially diminishing the fiber's fat-binding efficiency (some adaptation in gut flora or absorption might occur, as seen with other fibers). Additionally, while the cactus fiber clearly enhances weight loss when combined with a calorie-controlled diet, its effectiveness outside of a structured diet context (e.g., in a free-living population without dietary guidance) is less certain. The product should be viewed as a support for, not a substitute for, healthy eating and exercise.

Looking ahead, further research could explore combining OFI fiber with other weight-management strategies (such as other types of fiber or different supplements) to see if there is an additive benefit. Moreover, investigations into metabolic effects beyond weight – for instance, does long-term cactus fiber improve cholesterol profiles or glycemic control in obese patients? Some hints of these benefits exist (Godard et al., 2010; Frati-Munari et al., 1989), but modern well-powered trials would help confirm additional health benefits.

In conclusion, fiber derived from *Opuntia ficus-indica* (prickly pear cactus) has emerged as a credible, evidence-based supplement for weight management. It can modestly enhance weight loss and aid weight maintenance by binding dietary fats and increasing their excretion, all with a strong safety and tolerability record. While it is not a magic bullet, when integrated into a comprehensive weight loss program (dietary change, physical activity, behavioral support), OFI cactus fiber can contribute to achieving and sustaining healthier body weight. This natural approach addresses a key aspect of our diets – excessive fat intake – and does so in a way that is gentle on the system compared to pharmaceutical alternatives. For individuals seeking a scientifically supported supplement to assist in weight control, *Opuntia ficus-indica* fiber is a noteworthy option.

Efficacy of Non-Branded *Opuntia ficus-indica* Fiber Extracts

Most human trials on *Opuntia ficus-indica* (OFI) for weight management have used proprietary, standardized cactus fiber complexes such as NeOpuntia and Litramine (Grube et al., 2013; Uebelhack et al., 2014). These branded supplements are essentially dehydrated OFI cladode (prickly pear pad) extracts, sometimes blended with additional fibers to optimize their soluble fiber content and fat-binding capacity. For example, Litramine (IQP-G-002AS) is an OFI fiber complex enriched with acacia gum (a soluble fiber) and co-processed with cyclodextrin, giving it a high lipophilic binding activity (Grube et al., 2013). Likewise, NeOpuntia is a proprietary preparation of OFI cactus leaf fibers that has been tested in clinical settings, such as for metabolic syndrome and lipid management (Linarès et al., 2007). Both preparations have demonstrated the ability to bind dietary fats and modestly improve weight-related outcomes under controlled conditions (Grube et al., 2013; Uebelhack et al., 2014). These results, however, are not due to any unique magic of branding, but rather to the standardized fiber content and quality in the product.

Crucially, the fat-binding and weight-modulating effects of OFI supplements are believed to arise from the intrinsic properties of the cactus's dietary fiber itself. The cladodes of *O. ficus-indica* are naturally rich in fiber (primarily cellulose, hemicellulose, lignin, pectin, and mucilaginous polysaccharides) (Hernández-Urbiola et al., 2011). This fiber composition confers notable physicochemical capabilities, including high water-holding and lipid-binding capacity. In vitro analyses show that powdered OFI cladode fiber exhibits excellent water-binding and fat-absorption capacity, reflecting the ability of its fibers to

physically sequester lipids (Peña-Valdivia et al., 2012). Biochemically, the soluble fiber fraction (e.g., cactus pectins and mucilage) can bind bile acids and fats in the gastrointestinal tract—indeed, prickly pear pectin has been shown to act similarly to bile acid-binding resins in reducing cholesterol levels (Frati-Munari et al., 1983). In human trials, these fiber-rich cactus extracts significantly increased fat excretion: for instance, an OFI cactus fiber supplement led to a marked rise in fecal fat output compared to placebo, supporting the mechanism of fat binding and reduced fat absorption (Uebelhack et al., 2014). Such evidence underscores that the key functional mechanism—the adsorption of dietary fat by the fiber matrix, forming indigestible fat-fiber complexes—is an inherent property of the cactus fibers themselves (Grube et al., 2013; Uebelhack et al., 2014).

Accordingly, a high-quality non-branded OFI fiber extract, if properly prepared to preserve a similar soluble/insoluble fiber profile, should theoretically confer comparable benefits. The efficacy depends on the fiber's composition, fat-binding capacity, and delivered dose rather than the product's brand name. Standardized cactus fiber complexes are often characterized by their lipophilic binding ability (e.g., roughly 10 grams of fat bound per gram of fiber in the case of Litramine) (Grube et al., 2013), and an equivalently prepared OFI extract with similar fiber content can be expected to achieve analogous fat-binding ratios. In practice, ensuring an adequate dose of OFI fiber is critical—clinical studies have used on the order of 3–5 grams of cactus fiber extract per day (spread over meals) to obtain significant effects (Grube et al., 2013; Uebelhack et al., 2014). Thus, if a non-branded OFI fiber product delivers a comparable quantity of soluble and insoluble fiber (with high mucilage/pectin content) and has been processed in a way that retains its fat-binding functionality, it may produce similar weight-management outcomes as the branded formulations. In sum, the efficacy of cactus fiber for fat binding and weight control is fundamentally linked to the fiber's physicochemical properties—especially its ability to bind lipids and form unabsorbable complexes—and to appropriate dosing, rather than to any proprietary blend per se (Hernández-Urbiola et al., 2011; Grube et al., 2013; Uebelhack et al., 2014). This suggests that non-branded OFI fiber extracts of sufficient quality and standardization could offer the same physiological benefits, provided they match the fiber composition and fat-binding capacity demonstrated by the branded products in clinical trials (Peña-Valdivia et al., 2012).

Conflict of interest

The authors of this study are employees of Novogenia GmbH, a private commercial genetic laboratory that also funded this research. The scientific review focuses on the scientific evidence and basis of similar products being sold by the company.

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